The Particle Theory of Matter

Matter

There are thousands of different sorts of materials, all with their own special characteristics that make them suitable for particular uses. While they are all different, they all have one thing in common: they are all made of **matter**.

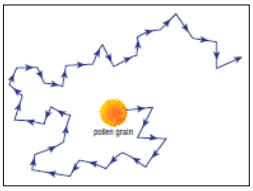
Matter is defined as anything that has **mass** (can be weighed) and **volume** (takes up space). Objects and materials are matter. Things that are not matter include time, heat and gravity. A space that does not contain any matter at all is called a vacuum. Outer space is very close to a perfect vacuum.

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In the particle model, matter is described as being made up of **very small particles which are constantly moving**. Scientists have built up this model by performing investigations and making observations of the behaviour of different types of matter.

Below are three example of evidence for the particle model.

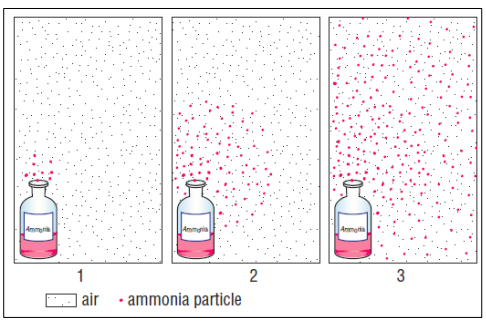
Brownian Motion



In 1827, Robert Brown first investigated pollen grains in water. He selected pollen because it is very light. Brown observed that the pollen grains were constantly moving around in a random manner. The particle model can be used to explain these observations. The particles that make up the water are constantly moving and bump into the pollen grains causing them to change direction. This type of movement was named Brownian motion, after Robert Brown.

Diffusion

Your nose is very sensitive to particles. Small particles move into your nose through the air and are detected by sensitive membranes. Being able to smell someone's perfume or something baking in the oven from a distance are examples of particles moving through the air into your nose. The movement of small particles through a substance, such as air or water, is called diffusion. The particles bump into each other, constantly changing directions and mixing until they are spread out fairly evenly. The figure on the left shows the diffusion of particles of ammonia in air.



Dissolving

Substances such as sugar can dissolve when put into water. Dissolving can be explained in terms of small particles moving through the water particles. This movement can be sped up by stirring, which forces the particles to move in the water faster.

Questions

1. How would you define matter?

2. What are some examples of things that are not matter?

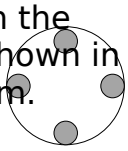
3. Why did Robert Brown use pollen grains in his experiment?

4. How does Brownian motion show us that particles exist?

5. Some strong perfume is placed at the front of a classroom. Eventually the perfume can be smelt at the back of the room. What is this process called and describe how it happens.

Particle Proof

Perform the following experiments and record your predictions and observations in the table.

	Prediction	**Observation**	**Explanation**
Fill a 250 mL beaker with water. Add 2 drops of food dye.			
Half fill a petri dish with water. Place four smarties in the water as shown in the diagram. 			
Half fill a petri dish with water.			

Put a small amount of copper sulphate on one side of the dish and a small amount of sodium bicarbonate on the other.			